

TS-TCN Component

Transaction-Sequence Temporal Convolutional Network

with fraud_attention Attribution

Technical Documentation for Research Paper Integration

DeepSentinel — Multi-Modal Financial Fraud Detection Platform

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1. Component Overview

The TS-TCN (Transaction-Sequence Temporal Convolutional Network) is the temporal-modality detector within the DeepSentinel platform. It classifies each financial transaction as fraudulent or legitimate by analysing the sequence of transactions that precede it, and it produces an intrinsic attribution output identifying which preceding transaction most influenced the decision.

Unlike the relational detector (GraphSAGE, Member 1) that models account-to-account connections, and the behavioural detector (Stratified VAE, Member 2) that models per-type reconstruction anomalies, the TS-TCN captures temporal escalation patterns — how fraud builds up across a sequence of transactions over time.

1.1 Role Within the Platform

- Input: a stream of financial transactions (PaySim schema during development; any mobile-money log in deployment).
- Output: a temporal risk score (0–1), a risk level (NORMAL / SUSPICIOUS / CRITICAL), and a forensic attribution payload.
- Downstream consumer: Member 4's Fusion Engine, which combines the TS-TCN output with the GraphSAGE and VAE outputs to produce the final fraud decision and LLM-generated report.
- Join key across all modules: `composite_id` formatted as `nameOrig_step`.

1.2 Core Idea

For every transaction under classification, the model assembles the 32 most recent preceding transactions across the entire transaction stream as a temporal context window. This produces one fully-labelled training sample per transaction with zero per-account history requirement, resolving the sequence-sparsity problem that renders account-level temporal models inapplicable in real-world mobile-money datasets where most accounts conduct very few transactions.

2. The Sub-Problem Addressed

Financial fraud detection in mobile-money environments faces three specific challenges that the TS-TCN component is designed to solve:

2.1 Frauds that build up across transactions are missed

Static threshold rules evaluate each transaction in isolation. They cannot detect fraud patterns that develop over a sequence — for example, progressively increasing drain ratios or the sequential use of empty destination accounts. In the PaySim dataset, the existing rule-based flag catches only 16 of 8,213 fraud cases.

2.2 Alerts lack a clear reason

Fraud alerts from black-box models arrive without explanation, leaving investigators unable to justify or act on them. Post-hoc explanation methods such as SHAP add significant inference latency, which is unacceptable for real-time deployment.

2.3 New accounts cannot be screened

Per-account sequential models (LSTM, per-account TCN) require long individual account histories. In mobile-money systems, most accounts conduct only one to three transactions, so these approaches are structurally inapplicable.

3. Novel Contributions

The TS-TCN component introduces four novelties, each addressing a specific limitation of existing approaches.

3.1 Novelty N1 — System-Wide Transaction-Stream Window

Instead of building sequences per account, the model constructs a system-wide window of the W ($=32$) most recent preceding transactions across the entire stream. This produces one fully-labelled training sample per transaction with zero per-account history requirement, so the model works even for brand-new accounts with no prior activity.

3.2 Novelty N2 — Dilated Causal TCN

A dilated causal Temporal Convolutional Network with dilation rates 1, 2, 4, and 8 captures both immediate-predecessor fraud signals and multi-transaction escalation patterns within a single architecture. Causal padding guarantees that no future-transaction information contaminates the classification decision, satisfying the hard requirement for real-time deployment.

3.3 Novelty N3 — fraud_attention Attribution Layer

A self-attention layer produces a weight vector over all window positions at inference time. The position with peak attention identifies the specific preceding transaction most influential to the fraud decision, and its complete feature vector is returned alongside its account identifier. This provides two complementary attribution outputs — which preceding transaction mattered and what features made it suspicious — from a single architecturally intrinsic mechanism that adds no inference latency, unlike post-hoc methods such as SHAP.

3.4 Novelty N4 — Data-Driven Type Risk Weight

The transaction-type risk weight feature is computed as the fraud-prevalence ratio per transaction type from the training partition only. This replaces arbitrary static encoding with a fully data-driven methodology that operators can recompute from their own labelled data in a single aggregation step, making the feature encoding universally portable across deployment environments.

4. Feature Engineering

Each transaction is represented by 10 engineered features (F1–F10). All features are universally computable from standard mobile-money transaction fields. Feature F6 (type risk weight) is derived from the training partition only, to prevent data leakage.

ID	Feature	Formula	Fraud Signal
F1	Drain ratio	$\min(\text{amount}/(\text{oldbalanceOrg}+1), 1.0)$	→1.0 when account fully drained
F2	Log amount	$\log_{1p}(\text{amount})$	Scale-normalised magnitude
F3	Post-transfer ratio	$\text{newbalanceOrig}/(\text{oldbalanceOrg}+1)$	→0.0 when account emptied
F4	Dest. was empty	$\text{float}(\text{oldbalanceDest}=0)$	Mule accounts start empty
F5	Dest. enrichment	$\min((\text{newDest}-\text{oldDest})/(\text{amount}+1), 2)$	Fraction of transfer that arrived
F6	Type risk weight	$\text{fraud_count_type} / \text{total_fraud} (\text{train})$	Data-driven prevalence by type
F7	Inv. dest ratio	$\log_{1p}(\text{oldDest})/(\log_{1p}(\text{newDest})+1)$	→0.0 for large transfer to empty
F8	Amt to orig	$\log_{1p}(\text{amount})/(\log_{1p}(\text{oldbalanceOrg})+1)$	→1.0 when amount exhausts balance
F9	Hour of day	$(\text{step}-1)\%24 / 23.0$	Fraud hour concentration
F10	Day of week	$((\text{step}-1)//24)\%7 / 6.0$	Weekly fraud patterns

Feature categories: Balance-flow (F1, F3, F8), Amount (F2), Destination-side (F4, F5, F7), Type/Temporal (F6, F9, F10).

5. Model Architecture

The TS-TCN takes a (32×10) window as input and outputs a fraud probability plus an attention-weight vector. It has four dilated causal TCN blocks, the fraud_attention layer, and a dense classification head.

5.1 Layer-by-Layer Specification

Layer	Configuration	Output Shape	Purpose
Input	32 transactions $\times$ 10 features	(batch, 32, 10)	Sequential context window
TCN Block 1	Conv1D(96, k=3, dilation=1, causal)	(batch, 32, 96)	Local 3-transaction patterns
TCN Block 2	Conv1D(96, k=3, dilation=2, causal)	(batch, 32, 96)	7-transaction context
TCN Block 3	Conv1D(96, k=3, dilation=4, causal)	(batch, 32, 96)	15-transaction context
TCN Block 4	Conv1D(96, k=3, dilation=8, causal)	(batch, 32, 96)	RF=61, full window coverage
fraud_attention	Self-attention, d_k=32	context + weights(32)	Transaction-level attribution
GlobalAvgPool1D	Compress temporal dimension	(batch, 96)	Fixed-length representation
Dense head	Dense(64,relu)+Drop(0.3)+Dense(1,sig)	(batch, 1)	Fraud probability 0–1

Total parameters: 218,945. **Loss:** BinaryFocalCrossentropy ($\gamma=2.0$). **Optimizer:** Adam ($\text{lr}=1\text{e-}3$). **Batch size:** 256.

5.2 Receptive Field

With kernel size 3 and dilations [1, 2, 4, 8] across 4 blocks (2 conv layers each), the receptive field is $RF = 1 + \sum (k-1) \cdot d = 61$. Since $61 > W=32$, every centre position can attend to the full predecessor history.

5.3 Causality Guarantee

Every Conv1D layer uses `padding='causal'`, so the output at time t depends only on inputs at positions $\leq t$. The window builder also snapshots the transaction buffer *before* appending the current transaction, giving a strict causal contract end-to-end — no future information ever leaks into the classification.

6. Baseline Comparison

Three baselines were evaluated under identical conditions to establish the performance gap the TS-TCN addresses. All use the same time-based train/test split (train: step ≤ 595 ; test: step > 595 , with 1,642 test fraud cases).

Model	Configuration	F1	Recall	Precision	AUC
B0 — Rule	TRANSFER & amount>200K	0.114	0.338	0.069	0.637
B1 — LogReg	liblinear, balanced	0.122	0.969	0.065	0.958
B2 — MLP	Dense(64)→Dense(32), focal $\gamma=2$	0.737	0.586	0.995	0.992
TS-TCN	4-block TCN + attention	pending	pending	pending	pending

Key finding: The MLP (B2) reaches 99.5% precision but only 58.6% recall — it misses 680 of 1,642 fraud cases because it evaluates each transaction in isolation without sequence context. Closing this recall gap is the primary objective of the TS-TCN. Final TS-TCN metrics are produced after full 30-epoch training.

7. Ablation Study Design

Four configurations isolate the contribution of each design choice. This study is scheduled for the evaluation phase.

ID	Configuration	Change	Hypothesis Tested
A1	W=32 + attention (primary)	Baseline config	All design choices combined
A2	W=32, no attention	Remove fraud_attention	Attention contribution to F1/AUC
A3	W=16 + attention	Reduce window to 16	Window-size contribution
A4	W=64 + attention	Increase window to 64	Diminishing returns beyond W=32

8. Output Contract for Fusion Engine

The TS-TCN delivers its output to Member 4's Fusion Engine via a POST request to `/api/v1/classify`. The response contains the risk score, risk level, current-transaction evidence, and the triggering-predecessor attribution. The full JSON schema is below.

8.1 JSON Response Schema

```
{
  "transaction_ref": {
    "nameOrig": "string (Account ID)",
    "step": "integer",
    "composite_id": "string (nameOrig_step)"
  },
  "temporal_risk_score": "float (0.0-1.0)",
  "risk_level": "NORMAL | SUSPICIOUS | CRITICAL",
  "detection_method": "TS-TCN",
  "modality": "temporal_sequence",
  "evidence": {
    "current_transaction": {
      "type", "amount", "drain_ratio",
      "post_transfer_ratio", "dest_was_empty",
      "type_risk", "hour_of_day",
      "fraud_signal_summary": "string (LLM anchor)"
    }
  },
  "triggering_predecessor": {
    "nameOrig", "composite_id", "attention_weight",
    "offset_from_current": "integer (e.g. -3)",
    "features": { "type", "amount", "drain_ratio",
      "post_transfer_ratio", "dest_was_empty", "type_risk" },
    "predecessor_signal": "string (forensic reason)"
  }
}
```

8.2 Critical Fields for the LLM Report

Field	Purpose
composite_id	Primary key linking to GraphSAGE and VAE outputs
fraud_signal_summary	Behavioural-reasoning anchor for the LLM report
triggering_predecessor	Forensic context — makes TS-TCN unique
attention_weight	Confidence in the chosen predecessor

Performance requirement: response returned within 500 ms for near-real-time processing.

9. Implementation Status

Stage	Component	Status
1	Feature engineering (F1–F10)	Complete
2	Baseline evaluation (B0, B1, B2)	Complete
3	System-wide window builder (6.24M windows)	Complete
4	TCN architecture + fraud_attention	Built; full training in progress
5	Ablation study (A1–A4)	Scheduled
6	Final evaluation + threshold tuning	Scheduled
7	Visualisations	Scheduled
8	FastAPI deployment endpoint	Scheduled

For the research paper, Sections 1–8 provide the complete methodology, architecture, and integration contract. Final quantitative results (TS-TCN metrics, ablation outcomes, ROC curves) will be supplied once full training and evaluation are complete.